

F360Core TrackerLib Software Complexity Report

Commit: 3f2a4363762fdf99f77b637634712b625c31439b

Introduction

The Software Complexity Report for the TrackerLib portion of the F360Core project is generated by Coverity. The tool supports C and C++, analyses code using its own checkers as well as MISRA checkers, and provides complexity statistics. The server responsible for continuous analysis is located at the APTIV branch in Gothenburg, Sweden and it is running Coverity Analysis version 2018.06.

Scope

This document applies to the F360Core Radar Tracker Library (TBD acronym) Roster number: F360CORE_00

The SW consists of modules which have different origin. In general each module will fall into one of the following categories:

- Third party modules directly integrated (in their original format/state) in the project.
- Third party code that are affected by our parameter configuration upon generation.
- Modules which are developed in the project by APTIV.

There are two possible approaches for generating the complexity report:

- Include all sources into the measurements including tools and third-party code which gives a more complete picture of the complexity of the whole SW. Drawback of this approach is that it shows call stacks and other information which is affected by parts that the project is unable to affect and makes it hard to detect APTIV implementation issues.
- The second approach is to only include modules which APTIV has implemented and thereby get feedback about possible issues introduced by the current project only.

Current report has been produced as the result of the first approach so that all potential risks can be identified. Issues which are not relevant will be excluded from the report during iterative review process supported by the utilized tools (Coverity).

Analyzed builds

Following builds are analyzed by Coverity:

- **F360TrackerLib** which is a library to be used when building Radar Tracker applications for radar sensors used with F360 systems.

Additionally there is a **Combined view** which provides results for all the above builds combined.

Lines of Code

Lines of Code metric is calculated as sum of lines in all files reported by Coverity as being part of a given build. If given line is compiled independently during builds for different cores it is counted each time in given build perspective, but in **Combined view** it is counted only once. That's why LOC in **Combined view** is not simply a sum of LOC from other builds.

build name	lines of code	comment lines	blank lines	total lines	files
Combined view	48700	45967	8854	103521	698
F360CORE_00_Track	48700	45967	8854	103521	698

Big files

File size is calculated as a sum of the following metrics reported by Coverity:

- lines of code
- empty (blank) lines
- comment lines

Summary

Table below presents how many files has more lines than the given thresholds.

build name	>10000	>5000	>2000	>1000
Combined view	0	0	0	12
F360CORE_00_TrackerLib	0	0	0	12

Details

Tables below presents files with high number of lines. Files longer than 5000 lines are marked RED.

Combined view

file name	lines of code	blank lines	comment lines	total
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/measurement_update_tracks/source/f360_pseudo_position_estimation.cpp	794	139	928	1861
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/common/source/f360_math_func.cpp	723	130	875	1728
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/f360_tracker.cpp	835	131	658	1624
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/trailer_detector/source/f360_tl_run.cpp	938	144	456	1538
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/include/f360_math_func.h	652	102	528	1282
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/detection_to_track_association/source/f360_detection_association_support_functions.cpp	492	85	509	1086
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/inputs_preprocessing/source/f360_sensor_capability.cpp	379	67	630	1076
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/initialize_tracks/source/f360_nees_cfmi_try_to_init.cpp	439	90	533	1062
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/track_downselection/source/f360_track_downselection_internal_functions.cpp	395	67	568	1030
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/common/source/f360_update_object_reference_point_support_functions.cpp	422	70	529	1021

F360CORE_00_TrackerLib

file name	lines of code	blank lines	comment lines	total
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/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/ F360Core/sw/F360TrackerLib/Fusion360/measurement_update_tracks/source/ f360_pseudo_position_estimation.cpp	794	139	928	1861
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/common/source/f360_math_func.cpp	723	130	875	1728
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/f360_tracker.cpp	835	131	658	1624
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/trailer_detector/source/f360_tl_run.cpp	938	144	456	1538
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/include/f360_math_func.h	652	102	528	1282
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/ F360Core/sw/F360TrackerLib/Fusion360/detection_to_track_association/source/ f360_detection_association_support_functions.cpp	492	85	509	1086
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/inputs_preprocessing/source/f360_sensor_capability.cpp	379	67	630	1076
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/initialize_tracks/source/f360_nees_cfmi_try_to_init.cpp	439	90	533	1062
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/ F360Core/sw/F360TrackerLib/Fusion360/track_downselection/source/ f360_track_downselection_internal_functions.cpp	395	67	568	1030
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/ modules/F360Core/sw/F360TrackerLib/Fusion360/common/source/ f360_update_object_reference_point_support_functions.cpp	422	70	529	1021

High comment ratio

Comment ratio is calculated out of the following metrics reported by Coverity:

- lines of code
- empty (blank) lines
- comment lines

Summary

Table below presents how many files that have a higher comment ratio than the given thresholds.

build name	>95%	>90%	>85%	>80%
Combined view	0	0	0	3
F360CORE_00_TrackerLib	0	0	0	3

Details

Tables below presents files with high comment ratio. Files with ratio higher than 80% are marked RED.

Combined view

file name	lines of code	blank lines	comment lines	comment ratio
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/static_environment/source/f360_longi_stat_cluster.cpp	12	4	70	81.40%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/initialize_tracks/source/f360_calc_object_init_ep.cpp	14	4	62	77.50%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/common/include/f360_get_reference_point_para_side.h	9	2	33	75.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/common/include/f360_get_reference_point_orth_side.h	9	2	33	75.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/common/source/f360_convert_tcs_posn_to_vcs_posn.cpp	20	0	57	74.03%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/common/source/f360_calculate_curvi_position.cpp	14	5	54	73.97%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/SharedTrackerAPI/Types/f360_velocity.h	9	3	34	73.91%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/initialize_tracks/include/f360_nees_cfmi_minimization.h	11	1	34	73.91%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/common/source/f360_convert_vcs_posn_to_tcs_posn.cpp	20	0	56	73.68%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/common/source/f360_object_track.cpp	24	2	71	73.20%

F360CORE_00_TrackerLib

file name	lines of code	blank lines	comment lines	comment ratio
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/static_environment/source/f360_longi_stat_cluster.cpp	12	4	70	81.40%

/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/initialize_tracks/source/f360_calc_object_init_ep.cpp	14	4	62	77.50%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/common/include/f360_get_reference_point_para_side.h	9	2	33	75.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/common/include/f360_get_reference_point_orth_side.h	9	2	33	75.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/common/source/f360_convert_tcs_posn_to_vcs_posn.cpp	20	0	57	74.03%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/common/source/f360_calculate_curvi_position.cpp	14	5	54	73.97%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/SharedTrackerAPI/Types/f360_velocity.h	9	3	34	73.91%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/initialize_tracks/include/f360_nees_cfmi_minimization.h	11	1	34	73.91%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/common/source/f360_convert_vcs_posn_to_tcs_posn.cpp	20	0	56	73.68%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/common/source/f360_object_track.cpp	24	2	71	73.20%

Low comment ratio

Comment ratio is calculated out of the following metrics reported by Coverity:

- lines of code
- empty (blank) lines
- comment lines

Summary

Table below presents how many files that have a lower comment ratio than the given thresholds.

build name	<5%	<10%	<15%	<20%
Combined view	12	29	56	86
F360CORE_00_TrackerLib	12	29	56	86

Details

Tables below presents files with low comment ratio. Files with ratio lower than 20% are marked RED.

Combined view

file name	lines of code	blank lines	comment lines	comment ratio
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ocg/src/iface/ocg_occupancy_grid_types.h	41	8	0	0.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/SharedTrackerAPI/core/f360_get_wall_time.h	4	1	0	0.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/autocode_reuse/include/f360_matrix_vector_Init_real32_T.h	48	9	0	0.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/autocode_reuse/include/f360_linear_solver_data_types.h	20	5	0	0.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/SharedTrackerAPI/Logging/f360_internal_object.h	69	5	2	2.63%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Timing/f360_get_wall_time.cpp	28	2	1	3.23%
/Compiler/TASKING/TriCorev6.2r2/ctc/include/math.h	344	25	13	3.40%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/Fusion360/logging/source/f360_populate_internal_objects_log.cpp	239	7	10	3.91%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/SharedTrackerAPI/Logging/f360_internal_detection_history.h	44	5	2	3.92%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/SharedTrackerAPI/Logging/f360_internal_cluster.h	38	6	2	4.35%

F360CORE_00_TrackerLib

file name	lines of code	blank lines	comment lines	comment ratio
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ocg/src/iface/ocg_occupancy_grid_types.h	41	8	0	0.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/F360TrackerLib/SharedTrackerAPI/core/f360_get_wall_time.h	4	1	0	0.00%

/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/autocode_reuse/include/f360_matrix_vector_Init_real32_T.h	48	9	0	0.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/autocode_reuse/include/f360_linear_solver_data_types.h	20	5	0	0.00%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/SharedTrackerAPI/Logging/f360_internal_object.h	69	5	2	2.63%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Timing/f360_get_wall_time.cpp	28	2	1	3.23%
/Compiler/TASKING/TriCorev6.2r2/ctc/include/math.h	344	25	13	3.40%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/Fusion360/logging/source/f360_populate_internal_objects_log.cpp	239	7	10	3.91%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/SharedTrackerAPI/Logging/f360_internal_detection_history.h	44	5	2	3.92%
/JenkinsOT/workspace/60_Tracker_Coverity_F360Core_dev/modules/F360Core/sw/ F360TrackerLib/SharedTrackerAPI/Logging/f360_internal_cluster.h	38	6	2	4.35%

Cyclomatic Complexity (McCabe)

Cyclomatic complexity is a quantitative measure of the number of linearly independent paths through a program's source code. More details about this metric can be found at https://en.wikipedia.org/wiki/Cyclomatic_complexity

Summary

Table below presents how many functions that have a higher cyclomatic complexity than the given thresholds.

build name	>120	>60	>30	>15
Combined view	0	0	5	19
F360CORE_00_TrackerLib	0	0	5	19

Details

Tables below presents functions with high cyclomatic complexity. Functions with complexity greater than 15 are marked RED.

Combined view

Function name	File name	lines	blocks	cyclomatic complexity
f360_variant_A::Trailer_Detector_TL::Shrink_Trailer_Length(bool, int, int, float &, bool &)	f360_tl_run.cpp	174	102	58
f360_variant_A::Trailer_Detector_TL::Estimate_Trailer_Length_Peaks()	f360_tl_run.cpp	180	98	56
f360_variant_A::Sanity_Check(const rsppp_variant_A:::RSPP_Detection_List_Tag &, const f360_variant_A:::F360_Detection_Props_Tag (&)[9408], const f360_variant_A:::F360_Detection_Hist_Tag &, const f360_variant_A:::F360_Cluster_Tag (&)[2000], const f360_variant_A:::F360_Object_Track_T (&)[1000], const f360_variant_A:::F360_Tracker_Info_Tag &, f360_variant_A:::F360_TRKR_TIMING_INFO_T &)	f360_sanity_check.cpp	193	128	52
f360_variant_A::Trailer_Detector_TL::Extend_Trailer_Length(bool, int, int, int, float &)	f360_tl_run.cpp	106	57	44
f360_variant_A::F360_Tracker::Initialize_Tracker_State_From_Log(const f360_variant_A:::F360_Radar_Sensor_Tag (&)[10], const f360_variant_A:::F360_Host_Tag &, const f360_variant_A:::F360_Object_Log_Output_Tag &, const Tracker_Info_Log_Tag &, const F360_Host_Props_Log &, const F360_Static_Env_Poly_Log_Tag &, const Trailer_Detector_Log_Tag &, const f360_variant_A:::Tracker_Internal_Tag &, const F360_Internal_Trailer_Detector_Tag &) const	f360_tracker.cpp	181	163	38
f360_variant_A::Adjust_Overlapping_Confirmed_Tracks(const f360_variant_A:::F360_Calibrations_Tag &, const f360_variant_A:::F360_Tracker_Info_Tag &, f360_variant_A:::F360_Object_Track_T (&)[1000], f360_variant_A:::F360_TRKR_TIMING_INFO_T &)	f360_adjust_overlapping_confirmed_tracks.cpp	118	75	24

f360_variant_A::Mark_Object_Tracks_Next_To_Sensors (const f360_variant_A::F360_Calibrations_Tag &, const f360_variant_A::F360_Radar_Sensor_Tag (&)[10], const f360_variant_A::F360_Object_Track_T (&)[1000], const f360_variant_A::F360_Tracker_Info_Tag &, f360_variant_A::F360_Radar_Sensor_Props_Tag (&)[100], f360_variant_A::F360_TRKR_TIMING_INFO_T &)	f360_mark_object_tracks_next_t o_sensors.cpp	144	56	23
f360_variant_A::Trailer_Detector_TP::Process_Input (const f360_variant_A::F360_Host_Tag &, const rspp_variant_A::RSPP_Detection_List_Tag &, const f360_variant_A::F360_Detection_Props_Tag (&)[9408], const f360_variant_A::F360_Radar_Sensor_Tag (&)[10])	f360_tp_run.cpp	80	39	21
f360_variant_A::Populate_Objects_Data(f360_variant_A::F360_Object_Track_T (&)[1000], f360_variant_A::F360_Tracker_Info_Tag &, const F360_Object_Log_Tag (&)[1000], const f360_variant_A::F360_Calibrations_Tag &, const f360_variant_A::F360_Host_Tag &)	f360_populate_objects_log.cpp	211	92	21
f360_variant_A::Trailer_Detector_TL::Locate_Peaks()	f360_tl_run.cpp	118	69	20

F360CORE_00_TrackerLib

Function name	File name	lines	blocks	cyclomatic complexity
f360_variant_A::Trailer_Detector_TL::Shrink_Trailer_Length(bool, int, int, float &, bool &)	f360_tl_run.cpp	174	102	58
f360_variant_A::Trailer_Detector_TL::Estimate_Trailer_Length_Peaks()	f360_tl_run.cpp	180	98	56
f360_variant_A::Sanity_Check(const rspp_variant_A::RSPP_Detection_List_Tag &, const f360_variant_A::F360_Detection_Props_Tag (&)[9408], const f360_variant_A::F360_Detection_Hist_Tag &, const f360_variant_A::F360_Cluster_Tag (&)[2000], const f360_variant_A::F360_Object_Track_T (&)[1000], const f360_variant_A::F360_Tracker_Info_Tag &, f360_variant_A::F360_TRKR_TIMING_INFO_T &)	f360_sanity_check.cpp	193	128	52
f360_variant_A::Trailer_Detector_TL::Extend_Trailer_Length(bool, int, int, int, float &)	f360_tl_run.cpp	106	57	44
f360_variant_A::F360_Tracker::Initialize_Tracker_State_From_Log(const f360_variant_A::F360_Radar_Sensor_Tag (&)[10], const f360_variant_A::F360_Host_Tag &, const f360_variant_A::F360_Object_Log_Output_Tag &, const Tracker_Info_Log_Tag &, const F360_Host_Props_Log &, const F360_Static_Env_Poly_Log_Tag &, const Trailer_Detector_Log_Tag &, const f360_variant_A::Tracker_Internal_Tag &, const F360_Internal_Trailer_Detector_Tag &) const	f360_tracker.cpp	181	163	38
f360_variant_A::Adjust_Overlapping_Confirmed_Tracks(const f360_variant_A::F360_Calibrations_Tag &, const f360_variant_A::F360_Tracker_Info_Tag &, f360_variant_A::F360_Object_Track_T (&)[1000], f360_variant_A::F360_TRKR_TIMING_INFO_T &)	f360_adjust_overlapping_confirmed_tracks.cpp	118	75	24
f360_variant_A::Mark_Object_Tracks_Next_To_Sensors (const f360_variant_A::F360_Calibrations_Tag &, const f360_variant_A::F360_Radar_Sensor_Tag (&)[10], const f360_variant_A::F360_Object_Track_T (&)[1000])	f360_mark_object_tracks_next_t o_sensors.cpp	144	56	23

[0], const f360_variant_A::F360_Tracker_Info_Tag & f360_variant_A::F360_Radar_Sensor_Props_Tag (&)[1 0], f360_variant_A::F360_TRKR_TIMING_INFO_T & f360_variant_A::Trailer_Detector_TP::Process_Input (const f360_variant_A::F360_Host_Tag &, const rspp _variant_A::RSPP_Detection_List_Tag &, const f360_ variant_A::F360_Detection_Props_Tag (&)[9408], con st f360_variant_A::F360_Radar_Sensor_Tag (&)[10]) f360_variant_A::Populate_Objects_Data(f360_variant _A::F360_Object_Track_T (&)[1000], f360_variant_A: :F360_Tracker_Info_Tag &, const F360_Object_Log_Ta g (&)[1000], const f360_variant_A::F360_Calibratio ns_Tag &, const f360_variant_A::F360_Host_Tag &) f360_variant_A::Trailer_Detector_TL::Locate_Peaks()	f360_tp_run.cpp	80	39	21
	f360_populate_objects_log.cpp	211	92	21
	f360_tl_run.cpp	118	69	20

Halstead Effort

Halstead Effort is a software metric that is based on number of operands and operators. More details about this metric can be found at https://en.wikipedia.org/wiki/Halstead_complexity_measures

Summary

Table below presents how many functions that have a higher Halstead effort than the given thresholds.

build name	>5000000	>500000	>50000	>5000
Combined view	0	0	3	18
F360CORE_00_TrackerLib	0	0	3	18

Details

Tables below presents functions with high Halstead effort. Functions with Halstead effort greater than 5000 are marked RED.

Combined view

Function name	File name	operands	operations	Halstead effort
f360_variant_A::Trailer_Detector_TL::Shrink_Trailer_Length(bool, int, int, float &, bool &)	f360_tl_run.cpp	58	21	55310.0
f360_variant_A::Trailer_Detector_TL::Estimate_Trailer_Length_Peaks()	f360_tl_run.cpp	69	19	60799.3
f360_variant_A::Sanity_Check(const rspp_variant_A::RSPP_Detection_List_Tag &, const f360_variant_A::F360_Detection_Props_Tag (&)[9408], const f360_variant_A::F360_Detection_Hist_Tag &, const f360_variant_A::F360_Cluster_Tag (&)[2000], const f360_variant_A::F360_Object_Track_T (&)[1000], const f360_variant_A::F360_Tracker_Info_Tag &, f360_variant_A::F360_TRKR_TIMING_INFO_T &)	f360_sanity_check.cpp	54	19	29671.7
f360_variant_A::Trailer_Detector_TL::Extend_Trailer_Length(bool, int, int, int, float &)	f360_tl_run.cpp	44	17	32450.5
f360_variant_A::F360_Tracker::Initialize_Tracker_State_From_Log(const f360_variant_A::F360_Radar_Sensor_Tag (&)[10], const f360_variant_A::F360_Host_Tag &, const f360_variant_A::F360_Object_Log_Output_Tag &, const Tracker_Info_Log_Tag &, const F360_Host_Props_Log &, const F360_Static_Env_Poly_Log_Tag &, const Trailer_Detector_Log_Tag &, const f360_variant_A::Tracker_Internal_Tag &, const F360_Internal_Trailer_Detector_Tag &) const	f360_tracker.cpp	89	14	47862.1
f360_variant_A::Adjust_Overlapping_Confirmed_Tracks(const f360_variant_A::F360_Calibrations_Tag &, const f360_variant_A::F360_Tracker_Info_Tag &, f360_variant_A::F360_Object_Track_T (&)[1000], f360_variant_A::F360_TRKR_TIMING_INFO_T &)	f360_adjust_overlapping_confirmed_tracks.cpp	47	20	21184.7
f360_variant_A::Mark_Object_Tracks_Next_To_Sensors(const f360_variant_A::F360_Calibrations_Tag &, co	f360_mark_object_tracks_next_to_sensors.cpp	57	18	25923.1

<pre> nst f360_variant_A::F360_Radar_Sensor_Tag (&)[10], const f360_variant_A::F360_Object_Track_T (&)[100 0], const f360_variant_A::F360_Tracker_Info_Tag &, f360_variant_A::F360_Radar_Sensor_Props_Tag (&)[1 0], f360_variant_A::F360_TRKR_TIMING_INFO_T &) f360_variant_A::Trailer_Detector_TP::Process_Input (const f360_variant_A::F360_Host_Tag &, const rspp _variant_A::RSPP_Detection_List_Tag &, const f360_ variant_A::F360_Detection_Props_Tag (&)[9408], con st f360_variant_A::F360_Radar_Sensor_Tag (&)[10]) f360_variant_A::Populate_Objects_Data(f360_variant _A::F360_Object_Track_T (&)[1000], f360_variant_A: :F360_Tracker_Info_Tag &, const F360_Object_Log_Ta g (&)[1000], const f360_variant_A::F360_Calibratio ns_Tag &, const f360_variant_A::F360_Host_Tag &) f360_variant_A::Trailer_Detector_TL::Locate_Peaks() </pre>	f360_tp_run.cpp	34	19	11102.3
<pre> f360_variant_A::Populate_Objects_Data(f360_variant _A::F360_Object_Track_T (&)[1000], f360_variant_A: :F360_Tracker_Info_Tag &, const F360_Object_Log_Ta g (&)[1000], const f360_variant_A::F360_Calibratio ns_Tag &, const f360_variant_A::F360_Host_Tag &) f360_variant_A::Trailer_Detector_TL::Locate_Peaks() </pre>	f360_populate_objects_log.cpp	62	18	123190.0
<pre> f360_variant_A::Trailer_Detector_TL::Locate_Peaks() </pre>	f360_tl_run.cpp	36	17	29827.7

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Function name	File name	operands	operations	Halstead effort
f360_variant_A::Trailer_Detector_TL::Shrink_Traile r_Length(bool, int, int, float &, bool &)	f360_tl_run.cpp	58	21	55310.0
f360_variant_A::Trailer_Detector_TL::Estimate_Trai ler_Length_Peaks()	f360_tl_run.cpp	69	19	60799.3
f360_variant_A::Sanity_Check(const rspp_variant_A: :RSPP_Detection_List_Tag &, const f360_variant_A:: F360_Detection_Props_Tag (&)[9408], const f360_var iant_A::F360_Detection_Hist_Tag &, const f360_vari ant_A::F360_Cluster_Tag (&)[2000], const f360_vari ant_A::F360_Object_Track_T (&)[1000], const f360_v ariant_A::F360_Tracker_Info_Tag &, f360_variant_A: :F360_TRKR_TIMING_INFO_T &)	f360_sanity_check.cpp	54	19	29671.7
f360_variant_A::Trailer_Detector_TL::Extend_Traile r_Length(bool, int, int, int, float &)	f360_tl_run.cpp	44	17	32450.5
f360_variant_A::F360_Tracker::Initialize_Tracker_S tate_From_Log(const f360_variant_A::F360_Radar_Sen sor_Tag (&)[10], const f360_variant_A::F360_Host_T ag &, const f360_variant_A::F360_Object_Log_Output _Tag &, const Tracker_Info_Log_Tag &, const F360_H ost_Props_Log &, const F360_Static_Env_Poly_Log_Ta g &, const Trailer_Detector_Log_Tag &, const f360_ variant_A::Tracker_Internal_Tag &, const F360_Inte rnal_Trailer_Detector_Tag &) const	f360_tracker.cpp	89	14	47862.1
f360_variant_A::Adjust_Overlapping_Confirmed_Track s(const f360_variant_A::F360_Calibrations_Tag &, c onst f360_variant_A::F360_Tracker_Info_Tag &, f360 _variant_A::F360_Object_Track_T (&)[1000], f360_va riant_A::F360_TRKR_TIMING_INFO_T &)	f360_adjust_overlapping_confir med_tracks.cpp	47	20	21184.7
f360_variant_A::Mark_Object_Tracks_Next_To_Sensors (const f360_variant_A::F360_Calibrations_Tag &, co nst f360_variant_A::F360_Radar_Sensor_Tag (&)[10], const f360_variant_A::F360_Object_Track_T (&)[100 0], const f360_variant_A::F360_Tracker_Info_Tag &, f360_variant_A::F360_Radar_Sensor_Props_Tag (&)[1 0], f360_variant_A::F360_TRKR_TIMING_INFO_T &)	f360_mark_object_tracks_next_t o_sensors.cpp	57	18	25923.1

f360_variant_A::Trailer_Detector_TP::Process_Input (const f360_variant_A::F360_Host_Tag &, const rspp _variant_A::RSPP_Detection_List_Tag &, const f360_ variant_A::F360_Detection_Props_Tag (&)[9408], con st f360_variant_A::F360_Radar_Sensor_Tag (&)[10])	f360_tp_run.cpp	34	19	11102.3
f360_variant_A::Populate_Objects_Data(f360_variant _A::F360_Object_Track_T (&)[1000], f360_variant_A: :F360_Tracker_Info_Tag &, const F360_Object_Log_Ta g (&)[1000], const f360_variant_A::F360_Calibratio ns_Tag &, const f360_variant_A::F360_Host_Tag &)	f360_populate_objects_log.cpp	62	18	123190.0
f360_variant_A::Trailer_Detector_TL::Locate_Peaks()	f360_tl_run.cpp	36	17	29827.7